Grading Rubric – AI Agent Analysis Homework

Paper: A Platform-Based Incentive Mechanism for Autonomous Vehicle Crowdsensing

Everybody should present: No presentation, no grade (no free riders)

Total Points: 100

Important: Answers must be written in the student's own words. Responses that appear copied from external sources or generated without understanding will not receive full credit. Students must be able to explain and defend their answers during the in-class presentation.

Section 1 – Agent and Environment Analysis (Conceptual) – 30 points

Component	Points	Criteria
Agent type classification (MAI)	10	Correct identification with justification tied to system behavior
Environment properties classification	10	Correct labels with reasoning
Knowledge representation and decision logic	5	Understanding of expected utility, sensing plan, internal state
Autonomy and rationality discussion	5	Clear explanation of why AVs qualify as rational autonomous agents

Section 2 – Numerical and Analytical Questions – 45 points

This section carries the highest weight. Full credit requires step-by-step reasoning.

Component	Points	Criteria
Expected Utility calculation	15	Correct probability reasoning and weighted sum, steps shown
Capacity constraint reasoning	5	Logical step-by-step explanation
Sensing plan and competition reasoning	10	Clear understanding of how utility changes with more agents
Binomial probability calculation	5	Correct formula and substitution
Poisson-Binomial explanation	10	Clear explanation of why probabilities differ per agent

Section 3 – Complexity and Algorithmic Reasoning – 15 points

Component	Points	Criteria

Identification of combinatorial explosion	5	Recognition of exponential growth of trajectory combinations
Explanation of Nash equilibrium difficulty	5	Understanding of strategy space growth
How ATNE reduces complexity	5	Explanation of greedy local decisions and approximation

Section 4 – Conceptual AI Connections – 10 points

Component	Points	Criteria
Multi-agent rational agent explanation	5	Correct link to rational agent concepts
Reinforcement Learning comparison	5	Insight into how learning would change classification

Explanation Quality Adjustment (Applied to Whole Assignment)

Scores may be adjusted by up to 10 percent based on explanation quality.

Level	Adjustment	Description
Excellent clarity	+5%	Clear, original explanations demonstrating deep understanding
Acceptable	0%	Mostly clear but some shallow explanations
Poor or copied style	-5%	Generic wording or vague reasoning
Lack of understanding during defense	-10%	Student cannot explain reasoning when asked

Academic Integrity Requirement

Students must write answers in their own words, show intermediate steps for calculations, avoid copying directly from sources, and be prepared to explain all answers.

In-Class Presentation and Defense (Required)

Each student or group must give a 5 to 7 minutes presentation explaining:

1. The agent type and justification

2. One environment classification they found interesting

3. One numerical problem and its solution steps

4. One complexity challenge in the paper

Presentation Bonus (up to 10 points added):

Criteria	Points
Clear explanation of agent type	3
Correct explanation of one numerical solution	3
Understanding of complexity issue	2
Ability to answer follow-up questions	2